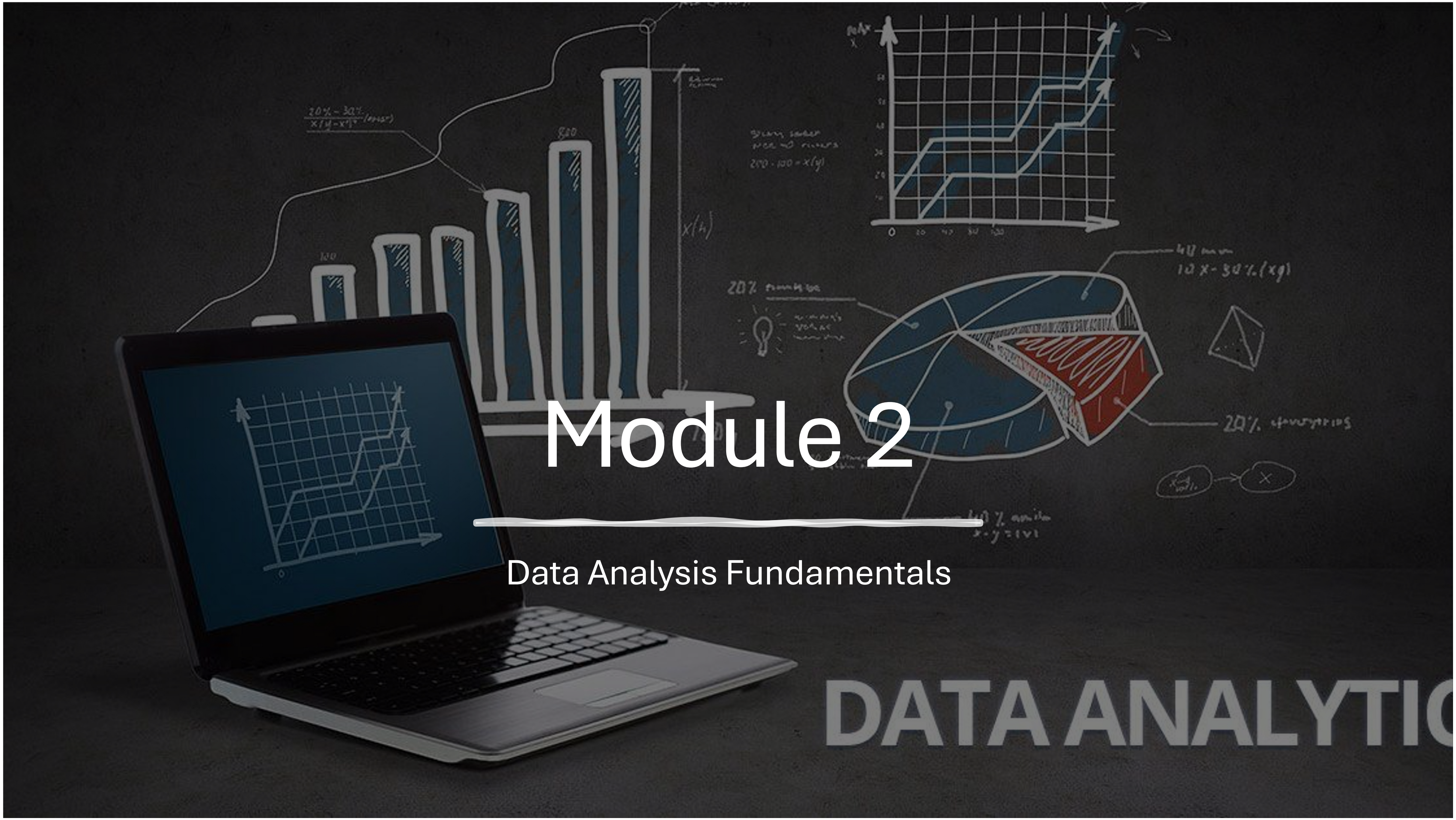
K-Youth Training Program

Data Analytics for business growth and sustainability in the workforce

Module 2

Data Analysis Fundamentals



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- 3. Storing and retrieving data
- 4. Data Quality and Cleaning Techniques
- 5. Data Transformation
- 6. Data Integration and ETL Processes
- 7. Python Basics for Non-IT Students
- a. Introduction to Python and Its Uses in Data Analytics
- b. Setting Up Python Environment
- c. Basic Python Syntax and Operations
- d. Data Structures: Lists, Tuples, Dictionaries
- e. Importing and Using Libraries (NumPy, Pandas)
- f. Basic Data Manipulation with Pandas

Data Quality and Cleaning Techniques

Understand the data

- Profile dataset to understand structure
- Check the data types
- Check missing values
- Look for inconsistencies

Duplicate handling

- Identify key columns
- Check for duplicates
- Review impact to the outcomes

Outliers

- Apply statistical methods
- Remove outliers with business rules
- Cap values based on threshold

Data Inconsistencies

- Standardize values
- Use reference tables for validation
- Domain based

Handle missing data

- Infer missing values using statistical method where possible (mean, median etc.)
- Drop rows or columns
- Domain knowledge

Data type handling

- Review columns and identify best data type for the data
- Update inconsistent data
- Used for consistency of numbers, dates, etc.

Standardizing and Normalization

- Uniform formats for data types in the dataset
- Scale data where required

Check data quality

- Adherence to business rules and constraints
- Define validation rules
- Check constraints

Common tools used

Process of data cleansing and cleaning can be divided into **automatic** and **manual** categories.

Automatic tools generally incorporate the features to manage an entire pipeline while manual tools like Excel can be used with some degree of automation.

Open source tools

- Python
- Pandas, NumPy, scikit-learn
- OpenRefine

Commercial tools

- Talend
- Alteryx
- Azure AI

Exercise

Use the Financial Excel file

- Standardize formats
- Values
- Update and reimports

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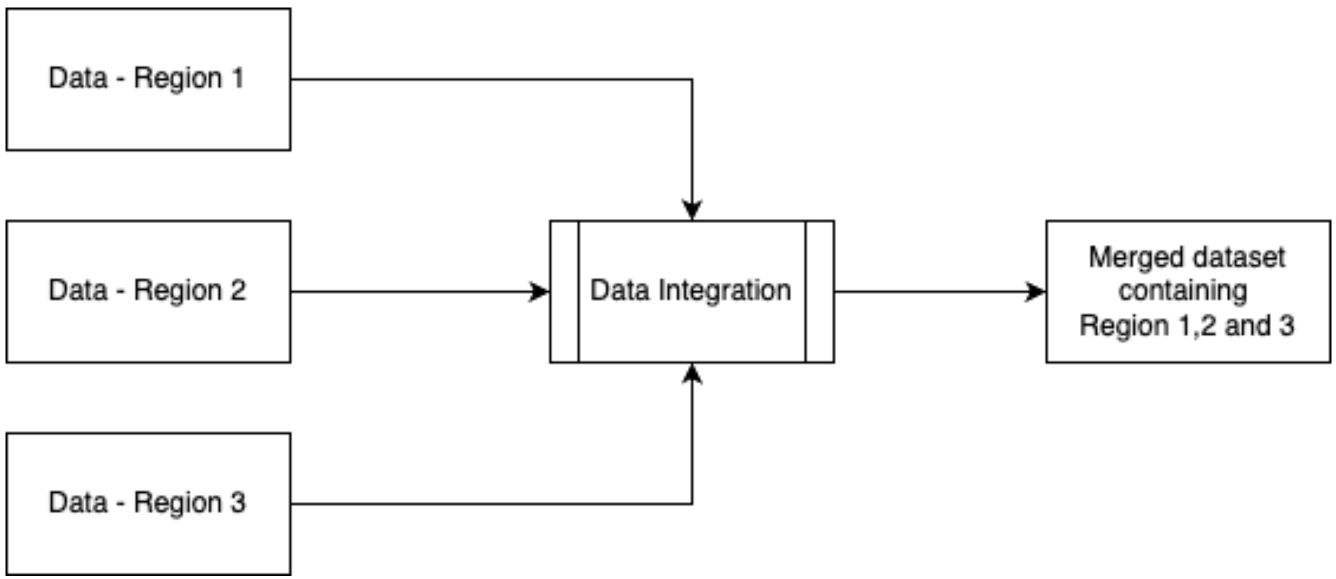
Data Transformation

After **data cleansing and quality assurance**, data transformation is the next critical step to prepare the data for analysis. Transformation involves converting the cleaned data into a format that is suitable for the intended analytical tasks.

- Data Integration
- Data Aggregation
- Feature Engineering
- Scaling and Normalization
- Pivot or Unpivoting
- Data Enrichment
- Time-Series Transformation
- Data Reduction
- Log Transformations
- Advanced Data Imputation
- Handling Imbalanced Data

Transformation – Data Integration

Data Integration - Combine data from multiple sources into a unified dataset



Transformation – Data Aggregation

Summarize data to a higher level of granularity

- Calculating metrics like sum, average, count, min, max, etc.
- Grouping data by specific categories or time periods.

Example

- Group Sales months and get average Sales

Year	Quarter	Month	Average of Total Sales
2022	Qtr 1	January	23158.53
2022	Qtr 1	February	23888.16
2022	Qtr 1	March	23232.98
2022	Qtr 2	April	21995.62
2022	Qtr 2	May	23253.79
2022	Qtr 2	June	23934.50
2022	Qtr 3	July	22222.79
2022	Qtr 3	August	23899.35
2022	Qtr 3	September	22170.21
2022	Qtr 4	October	25977.33
2022	Qtr 4	November	22559.24
2022	Qtr 4	December	22574.57
2023	Qtr 1	January	23772.14
2023	Qtr 1	February	24847.79
2023	Qtr 1	March	23672.06
2023	Qtr 2	April	24397.57
2023	Qtr 2	May	23879.71
2023	Qtr 2	June	23558.59
2023	Qtr 3	July	23243.42
2023	Qtr 3	August	24090.81
2023	Qtr 3	September	24146.98
2023	Qtr 4	October	24557.40
2023	Qtr 4	November	24267.90
2023	Qtr 4	December	24439.78
Total			**23955.07**

Transformation – Feature Engineering

Create new variables or features that improve the predictive power or interpretability of the data.

- **Binning**: Grouping continuous variables into categories (e.g., age ranges).
- **Encoding**: Converting categorical variables into numeric formats (e.g., one-hot encoding, label encoding).
- **Interaction Features**: Creating features that combine two or more existing variables (e.g., price_per_unit = total_cost / quantity).
- **Text Processing**: Extracting information like word count, sentiment, or keywords.

Transformation – Scaling and Normalization

- Ensure that numerical features are on a comparable scale.
- **Normalization**: Scaling data to a range (e.g., [0, 1]).
  - Example if using Min-Max on a dataset containing the values

    [50, 100, 150, 200, 250]
  - A scaled dataset will look like [0.0, 0.25, 0.5, 0.75, 1.0]

- **Standardization**: Transforming data to have a mean of 0 and a standard deviation of 1.

Transformation – Pivoting and Unpivoting

Reshape data for easier analysis or compatibility with analytical tools.

- **Pivoting**: Transform rows into columns for better representation of categories. For example you can pivot sales data to create a table where each region is a column

Date	Product	Sales
2024-01-01	Product A	100
2024-01-01	Product B	150
2024-01-02	Product A	120
2024-01-02	Product B	180

Date	Product A	Product B
2024-01-01	100	150
2024-01-02	120	180

- **Unpivoting**: Transform columns into rows to make data tidy.

Transformation – Data Enrichment

Data enrichment involves augmenting a dataset with additional information from external or internal sources to provide deeper insights, improve accuracy, or enable new analyses.

- Joining external data sources (e.g., appending weather data to sales records).
- Adding calculated fields based on domain knowledge.

Date	Store ID	Sales ($)
2024-01-01	101	1000
2024-01-01	102	1200
2024-01-02	101	1100
2024-01-02	102	1300

Date	City	Weather Condition	Temperature (°C)
2024-01-01	City A	Rainy	12
2024-01-01	City B	Sunny	25
2024-01-02	City A	Cloudy	15
2024-01-02	City B	Sunny	27

Date	Store ID	Sales ($)	Weather Condition	Temperature (°C)
2024-01-01	101	1000	Rainy	12
2024-01-01	102	1200	Sunny	25
2024-01-02	101	1100	Cloudy	15
2024-01-02	102	1300	Sunny	27

Transformation – Time-Series Transformation

Time series transformations are used to prepare, analyze, and forecast temporal data by modifying its structure or deriving new features.

These transformations often aim to uncover patterns, improve model performance, or meet the assumptions of specific algorithms.

- Creating lag features (e.g., sales_t-1).
- Rolling aggregates (e.g., moving averages, cumulative sums).
- Resampling or reindexing (e.g., from daily to weekly data).

Date	Sales
2024-01-01	100
2024-01-02	150
2024-01-03	200
2024-01-10	300

Week	Total Sales
2024-W1	450
2024-W2	300

Transformation – Data Reduction

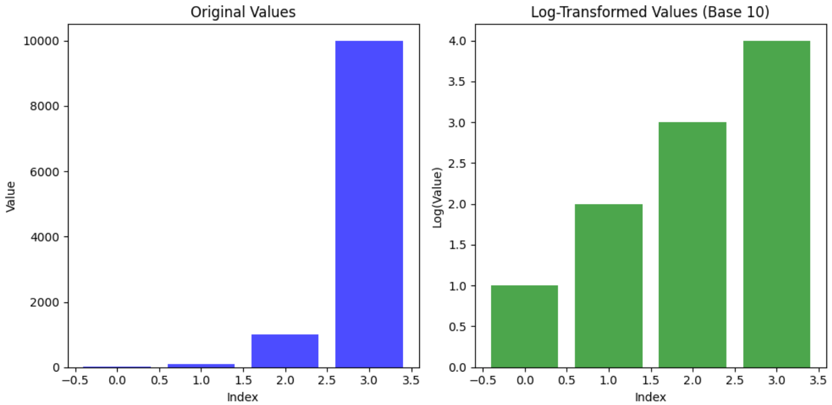
Data reduction involves transforming or compressing data to make it smaller, more manageable, and computationally efficient without significantly losing information. It is essential in large-scale analytics or when storage and processing resources are limited.

- Dimensionality reduction (e.g., PCA, t-SNE).
- Dropping irrelevant or redundant columns.
- Sampling a subset of the data for quicker analysis. Example from a data set of 1 million rows extract a random sample of 10,000 rows

Transformation – Log Transformations

Logarithmic transformations are used to reduce skewness in data, handle wide-ranging values, and make patterns in the data more apparent. They are particularly effective when the data contains exponential growth or right-skewed distributions.

- **Reduce Right Skewness**: When data has a long tail on the right.
- **Handle Outliers**: Mitigate the impact of extreme values.
- **Linearize Relationships**: Transform exponential growth into a linear relationship for easier modeling.



Transformation – Advanced Data Imputation

Data imputation is a critical step in data preprocessing where missing values are replaced with substituted values. Advanced imputation techniques are particularly effective for complex datasets where simple imputation (e.g., mean, median) might introduce bias.

- Impute values using advanced methods like KNN, regression, or interpolation

ID	Age	Salary	Experience
101	25	50000	2
102	NaN	52000	3
103	30	55000	NaN
104	27	NaN	4

ID	Age	Salary	Experience
101	25	50000	2
102	26	52000	3
103	30	55000	3.5
104	27	51000	4

Transformation – Handling Imbalanced Data

Handling imbalanced datasets is critical in data analytics, particularly for classification problems where one class has significantly fewer instances than the other(s). Below are techniques with Python examples to handle imbalanced datasets

- Oversampling (e.g., SMOTE),
- Undersampling, or using weighted algorithms.